

SkillSwap — Database Schema

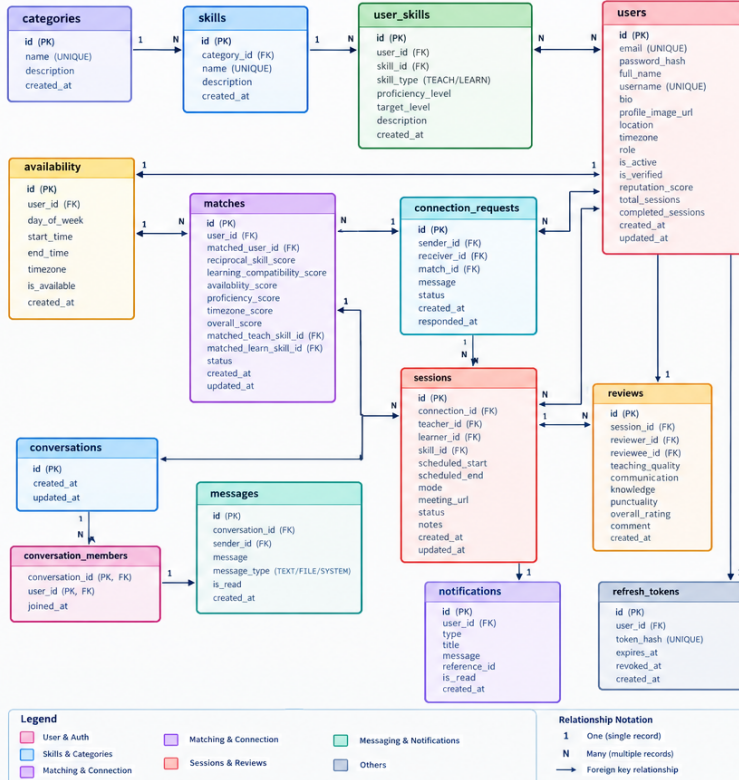
Entity-Relationship Diagram & PostgreSQL Data Model



Learn × Teach × Grow

Exchange skills. Build connections. Grow together.

ER Diagram



PostgreSQL Table Schemas



Relational Schema Overview

Core tables, keys, and relationships used by SkillSwap.

Table	Purpose	Primary Key	Important Foreign Keys / Constraints
users	User accounts and profiles	PK: id	email, username UNIQUE
categories	Skill categories	PK: id	name UNIQUE
skills	Master skill catalog	PK: id	category_id → categories.id
user_skills	Skills a user teaches or wants to learn	PK: id	user_id, skill_id → users/skills
availability	Weekly availability windows	PK: id	user_id → users.id
matches	Compatibility scores between users	PK: id	user_id, matched_user_id → users.id
connection_requests	Requests created from matches	PK: id	sender/receiver → users.id
sessions	Scheduled skill-exchange sessions	PK: id	connection_id, users, skill
reviews	Post-session reputation feedback	PK: id	session_id, reviewer/reviewee
conversations	Chat containers	PK: id	1:N conversation members
conversation_members	Users participating in conversations	PK: conversation_id + user_id	→ conversations/users
messages	Chat messages	PK: id	conversation_id, sender_id
notifications	User notifications	PK: id	user_id → users.id
refresh_tokens	Rotating refresh-token records	PK: id	user_id → users.id

Key relationship flow

Users → User Skills → Skills → Categories | Users → Matches → Connection Requests → Sessions → Reviews | Users → Conversations → Messages | Users → Notifications / Refresh Tokens

PostgreSQL Schema — Core DDL

Production-oriented structure using UUID primary keys and explicit foreign-key relationships.

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;

CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email VARCHAR(255) UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,
  full_name VARCHAR(150) NOT NULL,
  username VARCHAR(50) UNIQUE NOT NULL,
  bio TEXT,
  profile_image_url TEXT,
  location VARCHAR(150),
  timezone VARCHAR(100),
  role VARCHAR(30) DEFAULT 'USER',
  is_active BOOLEAN DEFAULT TRUE,
  is_verified BOOLEAN DEFAULT FALSE,
  reputation_score DECIMAL(3,2) DEFAULT 0,
  total_sessions INT DEFAULT 0,
  completed_sessions INT DEFAULT 0,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE categories (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(100) UNIQUE NOT NULL,
  description TEXT,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE skills (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  category_id UUID REFERENCES categories(id),
  name VARCHAR(100) UNIQUE NOT NULL,
  description TEXT,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE user_skills (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  skill_id UUID NOT NULL REFERENCES skills(id) ON DELETE CASCADE,
  skill_type VARCHAR(10) NOT NULL CHECK (skill_type IN ('TEACH','LEARN')),
  proficiency_level VARCHAR(20),
  target_level VARCHAR(20),
  description TEXT,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  UNIQUE(user_id, skill_id, skill_type)
);

CREATE TABLE availability (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  day_of_week SMALLINT NOT NULL CHECK (day_of_week BETWEEN 0 AND 6),
  start_time TIME NOT NULL,
  end_time TIME NOT NULL,
  timezone VARCHAR(100),
  is_available BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE matches (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  matched_user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  reciprocal_skill_score DECIMAL(5,2),
  learning_compatibility_score DECIMAL(5,2),
  availability_score DECIMAL(5,2),
  proficiency_score DECIMAL(5,2),
  timezone_score DECIMAL(5,2),
  overall_score DECIMAL(5,2),
  matched_teach_skill_id UUID REFERENCES skills(id),
  matched_learn_skill_id UUID REFERENCES skills(id),
  status VARCHAR(20) DEFAULT 'ACTIVE',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE connection_requests (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  sender_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  receiver_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  match_id UUID REFERENCES matches(id),
  message TEXT,
  status VARCHAR(20) DEFAULT 'PENDING',
```

```

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
responded_at TIMESTAMP
);

CREATE TABLE sessions (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
connection_id UUID REFERENCES connection_requests(id),
teacher_id UUID NOT NULL REFERENCES users(id),
learner_id UUID NOT NULL REFERENCES users(id),
skill_id UUID NOT NULL REFERENCES skills(id),
scheduled_start TIMESTAMP NOT NULL,
scheduled_end TIMESTAMP NOT NULL,
mode VARCHAR(20),
meeting_url TEXT,
status VARCHAR(20) DEFAULT 'SCHEDULED',
notes TEXT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE reviews (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
session_id UUID NOT NULL REFERENCES sessions(id) ON DELETE CASCADE,
reviewer_id UUID NOT NULL REFERENCES users(id),
reviewee_id UUID NOT NULL REFERENCES users(id),
teaching_quality SMALLINT CHECK (teaching_quality BETWEEN 1 AND 5),
communication SMALLINT CHECK (communication BETWEEN 1 AND 5),
knowledge SMALLINT CHECK (knowledge BETWEEN 1 AND 5),
punctuality SMALLINT CHECK (punctuality BETWEEN 1 AND 5),
overall_rating DECIMAL(2,1),
comment TEXT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE conversations (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE conversation_members (
conversation_id UUID REFERENCES conversations(id) ON DELETE CASCADE,
user_id UUID REFERENCES users(id) ON DELETE CASCADE,
joined_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (conversation_id, user_id)
);

CREATE TABLE messages (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
conversation_id UUID NOT NULL REFERENCES conversations(id) ON DELETE CASCADE,
sender_id UUID NOT NULL REFERENCES users(id),
message TEXT NOT NULL,
message_type VARCHAR(20) DEFAULT 'TEXT',
is_read BOOLEAN DEFAULT FALSE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE notifications (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
type VARCHAR(50) NOT NULL,
title VARCHAR(200) NOT NULL,
message TEXT NOT NULL,
reference_id UUID,
is_read BOOLEAN DEFAULT FALSE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE refresh_tokens (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
token_hash TEXT UNIQUE NOT NULL,
expires_at TIMESTAMP NOT NULL,
revoked_at TIMESTAMP,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```